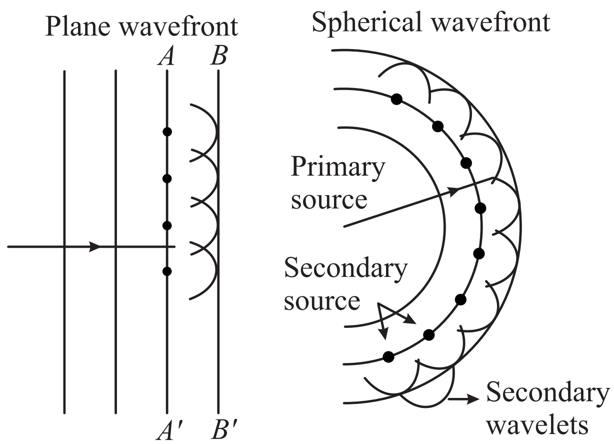
CHAPTER

10 Wave Optics

Huygen's Wave Theory

- Each point source of light is a center of disturbance from which waves are emitted in all directions. The locus of all the particles of the medium oscillating in the same phase at a given instant is called a wavefront.
- Each point on a wave front is a source of new disturbance, called secondary wavelets. These wavelets are spherical and travel with speed of light in that medium.
- The forward envelope of the secondary wavelets at any instant gives the position of the new wavefront.
- In homogeneous medium, the wave front is always perpendicular to the direction of wave propagation.



Coherent Sources

Two sources are coherent if and only if they produce waves of same frequency (and hence wavelength) and have a constant initial phase difference.

Incoherent sources

Two sources are said to be incoherent if they have different frequency or initial phase difference varies with time.

Interference : YDSE

- Resultant intensity for coherent sources

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi_0$$

- Resultant intensity for incoherent sources

$$I = I_1 + I_2$$

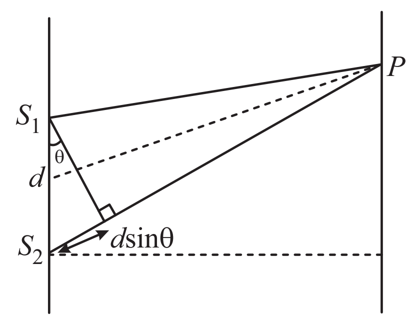
- Intensity $\propto$ width of slit $\propto$ (amplitude)^2

$$\Rightarrow \frac{I_1}{I_2} = \frac{W_1}{W_2} = \frac{A_1^2}{A_2^2}$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2$$

- Distance of nth bright fringe $y_n = \frac{n\lambda D}{d}$

  Path difference = $n\lambda$, where $n = 0, 1, 2, 3, ...$



- Distance of m^{th} dark fringe $y_m = \frac{(2m-1)\lambda D}{2d}$

  Path difference = $(2m - 1)\frac{\lambda}{2}$ where $m = 1, 2, 3,.....$

- Fringe width $\beta = \frac{\lambda D}{d}$
- Angular fringe width $\theta = \frac{\beta}{D} = \frac{\lambda}{d}$
- If a transparent sheet of refractive index μ and thickness t is introduced in one of the paths of interfering waves, optical path will become 'μt' instead of 't'. Entire fringe pattern shifts by $\frac{D\left[(\mu-1)t\right]}{d} = \frac{\beta}{\lambda}(\mu-1)t$ towards the side in which the thin sheet is introduced without any change in fringe width.

Diffraction

- In Fraunhoffer diffraction
  - For minima $a \sin\theta_n = n\lambda$
  - For maxima $a \sin\theta_n = (2n + 1)\frac{\lambda}{2}$

✦ Linear width of central maxima $W = \frac{2\lambda D}{a}$

✦ Angular width of central maxima $W_\theta = \frac{2\lambda}{a}$

Polarization

Brewster's law

$\mu = \tan\theta_p \Rightarrow \theta_p = \tan^{-1}\mu$, θ_p is polarization or Brewster's angle.

Here reflecting and refracting rays are perpendicular to each other.

Malus law

Malus law states that the intensity I of plane polarized that passes through an analyser varies as square of cosine of the angle between the plane of polariser and transmission axis of the analyser.

$I = I_0 \cos^2\theta$, $I_0 \rightarrow$ Intensity of incident polarized light